The Standard-Model floor: uniqueness of the minimal chiral matter content on a derived charge lattice

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Abstract

Within the sealed-records program (Parts I–II), we present the selection of Standard-Model matter content as a sequence of exhaustive, machine-verified searches under derived constraints — with every input named and graded. Derivations of this kind carry a well-known circularity risk (constraints read off the answer and fed back as selection principles); our discipline is to make every constraint's provenance explicit, and where a residue of target-calibration remains — it does, and we say where — to grade it as input rather than claim it as output. Results: (i) **hypercharge is the relative determinant character** of the reconstructed $U(3) \times U(2)$ fiber tower, $Y_6 = 3b - 2a$: with the $U(d)$ linkages $a \equiv \tau \pmod 3$, $b \equiv \sigma \pmod 2$ this renders the empirical $\mathbb{Z}_6$ congruence $2\tau + 3\sigma + Y_6 \equiv 0$ an *identity*, and the correspondence is sharp — of the thirty-six residue classes of determinant-character combinations exactly one yields the congruence; (ii) the selection *mechanism* is a charge-screening theorem: sealed, fermion-even, gauge-singlet records collapse superselection classes (a collapse lemma whose exclusion clauses are carried by univalence superselection and the centrality of the recorded gradings), the ledger's baryon–lepton seal at charge $(3, 2)$ screens exactly one direction, and a rank bound from the program's screen-$U(1)$ forbids more — the bare-baryon direction is blocked precisely by the fermionic-intertwiner exclusion; (iii) given only the representation shells, the Yukawa seam system plus the three Y-linear anomaly conditions has a one-dimensional kernel whose minimal integer point is exactly the Standard-Model hypercharge assignment — and the gauge-squared anomaly coefficients themselves are *lattice-grounded at abelianized scope* by a four-dimensional record index theorem (overlap index $= c^2 f_1 f_2$, exact in all sectors; $T(R)$ as summed Cartan indices; the full nonabelian rows remain imports pending instanton-sector receipts); (iv) the **floor theorem**: exhaustive search over the lattice — derived modulo the two named empirical inputs of (ii), which is what "derived" means everywhere in this paper — finds the Standard-Model generation as the unique minimal genuinely chiral anomaly-free content (15 Weyl fermions; nothing at ≤ 14; only the conjugate at 15), and a tower-exclusion search shows the $(3, 2)$ fiber tower has the strictly smallest chiral floor among alternatives, which either cost more than double or admit no floor at all in bounds. Falsifiable consequences include the absence of fractionally charged color singlets. The honest negatives and the named inputs are printed with equal prominence, and the input list is complete: the seam system fixes the hypercharge

ray but not the residue class (screening does); the minimality principle selecting the cheapest tower is a named input, as are the tower-uniformity premise of the exclusion comparison and the gauged-family premise of Section 6; the generation count is observational; **the existence of sealed fermionic records in the specific composite fibers needed to write the baryon–lepton seal (Q-type $(3, 2)$ and L-type $(1, 2)$ carriers) is an empirical input about nature's ledger, not an output** — it is the largest single input of the paper and Honest negative 3.5 discusses it; the surviving continuous charge functional (the screen-$U(1)$) and the weak-doublet scalar seam carrier (the Higgs representation) are likewise named inputs (Sections 3 and 7).

1. Introduction

Attempts to "derive the Standard Model" founder on two reefs: hidden circularity (constraints read off the answer and fed back as selection principles) and unfalsifiability (enough freedom to fit anything). This paper's discipline is aimed at both: every constraint used by the searches is either derived inside the program with receipts, or named as an input and graded; and the output is accompanied by falsifiers.

The structural inputs from Parts I–II: a fiber tower whose gauge structure is the *full* unitary group of each fiber (determinant slots included — the commutant reconstruction), univalence superselection, and the records-are-relations ontology. The new content here: the hypercharge derivation, its screening mechanism, the anomaly-row grounding, and the exhaustive floor searches.

2. Hypercharge as the relative determinant character

$U(d)$ representation theory links the determinant charge to the center: $a \equiv \tau \pmod 3$ for $U(3)$ (triality τ), $b \equiv \sigma \pmod 2$ for $U(2)$ (duality σ). Define hypercharge as a determinant-character combination $Y_6 = \alpha a + \beta b$.

Theorem 2.1 (equivalence, sharp). The empirical congruence $2\tau + 3\sigma + Y_6 \equiv 0 \pmod 6$ — satisfied by all six multiplets of the ν_R-completed Standard-Model generation — holds *identically in* (a, b) iff $(\alpha, \beta) \equiv (-2, 3) \pmod 6$, i.e. iff $Y_6 = 3b - 2a$ is the relative determinant character; then it is the identity $2a + 3b + (3b - 2a) = 6b$. Of the thirty-six residue classes, exactly one survives; every congruence-allowed charge triple is realized; and the minimal (a, b) representatives reproduce the Standard-Model table exactly ($Q(1, 1) \mapsto +1$, $L(0, -1) \mapsto -3$, $u^c(2, 0) \mapsto -4$, $d^c(-1, 0) \mapsto +2$, $e^c(0, 2) \mapsto +6$, $\nu^c(0, 0) \mapsto 0$).

Receipts: the 36-class scan and the zero-unrealized-triples check do real work; the $[-9, 9]^2$ identity scan is a sanity check of an algebraic identity and is labeled as such, not as evidence. Equivalently, at the global level, the gauge group is $S(U(3)\times U(2)) \cong (SU(3)\times SU(2)\times U(1))/\mathbb{Z}_6$ — the "true gauge group" structure documented by O'Raifeartaigh, Hucks, and Tong [1, 11, 2], here arrived at from the fiber side.

3. The mechanism: charge screening by sealed singlets

Why is hypercharge *that* character? The mechanism is a screening theorem with three clauses, each carried by a previously-established result:

Lemma 3.1 (collapse). A sealed, fermion-even, gauge-singlet record with charge displacement c collapses the superselection classes $[q]$ and $[q+c]$. *Exclusions*: fermionic intertwiners cannot collapse (univalence is superselected — Part II); gauge-charged ones cannot (they shift recorded labels); the centrality of the recorded gradings — by which we mean precisely that the $(-1)^F$ and determinant-charge gradings commute with every record-algebra element (the Part II superselection theorems; "code centrality" in earlier program vocabulary, retired here) — makes both statements one commutator computation (receipts: the even seal commutes exactly with the grading, the odd bare-baryon seal anticommutes exactly). *Membership*: sealed records are ledger morphisms by axiom R. *Collapse*: the GNS argument — the seal's nonvanishing action places a charge-$(q+c)$ module inside the charge-q representation.

Lemma 3.2 (the seal, stated fiber-theoretically — and its input content). In the fiber tower there exists an invariant tensor of type $\varepsilon^{(3)} \otimes \varepsilon^{(2)}$ coupling three color-fiber strands carrying weak-fiber indices to one bare weak-fiber strand (invariant-tensor dimensions 1 and 2, receipted); a sealed, fermion-even ($F = 4$) record built on it carries determinant charge exactly $(3, 2)$. **The input, stated where the lemma is used and in the abstract**: that nature's ledger contains *fermionic* records in those composite fibers — Q-type $(3, 2)$ and L-type $(1, 2)$ carriers — sufficient to instantiate this tensor. The familiar name of the resulting record is $QQQL$, and we are explicit that using it before the floor search outputs Q and L is not circular *only because* the lemma needs existence of the carriers, not their uniqueness or the rest of the generation — but that existence is itself an empirical fact, the single largest input of the paper.

Lemma 3.3 (rank bound; its premise graded). Screening two independent directions would annihilate every continuous charge functional. The premise that one continuous charge functional must survive — the program's screen-$U(1)$, whose derivation is corpus-bound [**P**] (Part I's tag for dependencies whose proofs live in the program corpus, not in this batch) — is **graded here as a named input co-equal with the seal**: it is, in effect, the requirement that a long-range abelian charge (electromagnetism) exists. Granting it, the screening lattice has rank ≤ 1, and containing the primitive $(3, 2)$ it is exactly $\mathbb{Z}(3, 2)$.

Theorem 3.4. The recordable charge data is (a, b) modulo $(3, 2)\mathbb{Z}$, whose complete invariants are exactly $(\tau, \sigma, Y_6 = 3b - 2a)$ — the $S(U(3) \times U(2))$ structure, hence Theorem 2.1's lattice, *derived modulo the graded inputs of Lemmas 3.2–3.3.* (Remark: Y_6 alone is already a complete invariant — $\tau \equiv Y_6 \bmod 3$ and $\sigma \equiv Y_6 \bmod 2$ — so the triple is stated for readability, not necessity.) Nonvanishing of the seal on arbitrary backgrounds holds by per-sector clustering (the transfer is self-adjoint by reflection positivity, so $T^n \to P_1$ needs no gap — an engine theorem valid at every rank; proof corpus-bound [**P**], flagged as a stated dependency of this batch), with a wrong-sector zero control showing the per-sector qualifier is exactly right.

Honest negative 3.5. The Yukawa-seam system (Section 4) does *not* select the residue class — the Standard-Model values also fit a wrong-class formula at residue level. The class selection rests on the screening mechanism; an alternative world whose ledger sealed a lepton-pair direction instead would have hypercharge proportional to the pure color determinant — consistent mathematics, excluded by *which seals nature's ledger contains*, an empirical fact we flag rather than bury.

4. The seam system and the anomaly rows, grounded

The input, named first. That hypercharge is fixed by anomaly cancellation *plus the Standard-Model Yukawa structure* is a known result, established by Geng–Marshak, Babu–Mohapatra, Foot et al., and Minahan–Ramond–Warner [3, 6, 7]; we claim no priority for the linear algebra of this section. The *existence of the four Yukawa seams and the Majorana seal* — i.e., which couplings the ledger contains — is itself structure: in the program it is carried by the records-are-relations axiom (every gauge-invariant fermion bilinear admits a sealing record), which is weaker than assuming the SM Yukawa matrix but is an input nonetheless, and we grade it as such. What this section adds beyond the classical result is (i) the provenance of the anomaly rows themselves (Theorem 4.1) and (ii) the division of labor established in Section 3: the seam system fixes the *ray*, screening fixes the *residue class*.

Given only the representation shells, impose the four Yukawa seams, the Majorana seal, and three Y-linear anomaly rows (gravitational, $SU(3)^2$–Y, $SU(2)^2$–Y): the system has rank 6 with a one-dimensional kernel whose minimal integer generator is $(y_Q, y_u, y_d, y_L, y_e, y_\nu \,|\, y_H) = (1, -4, 2, -3, 6, 0 \,|\, -3)$ — the Standard Model exactly. Without the gauge-squared rows the kernel is two-dimensional: they are load-bearing. Their provenance is *partially* upgraded by the following — precisely: the abelianized coefficients become lattice observables, while the full nonabelian rows remain continuum imports pending the instanton-sector receipt:

Theorem 4.1 (4d record index, abelianized scope). On the four-torus record lattice with overlap fermions [8] and $U(1)$ fluxes f_1, f_2 in two planes, the index equals $c^2 f_1 f_2$ — exact in six sectors including sign flips and charge two; with two independent $U(1)$s, the index is $q_a q_b f_1 f_2$; and Cartan-abelianized $SU(2)$ weight sums reproduce $T(R)\, m_1 m_2$ exactly (adjoint $+2.0$; fundamental at even fluxes $+2.0$ — the half-integer weights demanding even fluxes is the quotient structure of Section 2 reappearing from the lattice side). The quadratic anomaly coefficient, invisible to two-dimensional index theorems, is thereby a lattice observable, and the anomaly rows are sums of measured indices. The mathematical foundation — that the overlap construction carries an exact index on the lattice tied to the topological charge — is the Hasenfratz–Laliena–Niedermayer / Lüscher index theory [9]; our contribution is the *use*: measuring the gauge-squared coefficients as index data on record lattices, so the seam-system rows have lattice provenance instead of being continuum imports. (Full nonabelian backgrounds — instanton sectors — are the named next receipt.)

5. The floor theorems

Theorem 5.1 (uniqueness on the derived lattice; bounds stated). Exhaustive search over multiplets — color $\in \{1, 3, \bar{3}\}$, weak $\in \{1, 2\}$, hypercharges on the derived lattice with $|Y_6| \leq 12$ (i.e. $|Y| \leq 2NM$ for the $(3, 2)$ tower), at most five distinct multiplets, each with multiplicity, counted in Weyl fermions — under the anomaly predicates ($SU(3)^3$, $SU(3)^2$–Y, $SU(2)^2$–Y, Y^3, gravitational), the Witten condition [5] on pseudoreal doublets, and genuine chirality (no vector-like pairing of the full content): **zero** solutions below 15 Weyl fermions, and at 15 exactly the Standard-Model generation and its conjugate. Robustness: widening the hypercharge window and the representation zoo (color sextets and octets, weak triplets) at ≤ 16

Weyl admits no competitor below 16, with the first exotic alternative appearing at 16 Weyl in 6 multiplets. These are *search* results: their scope is exactly the stated pool and bounds, no more. The genre has prior art — systematic classifications of anomaly-free chiral matter (e.g. the Allanach–Gripaios–Tooby-Smith program [12]) — and the distinguishing feature here is not the search machinery but the *derived lattice* it runs on; we engage that literature as Section 4 engages the anomaly-quantization literature.

Theorem 5.2 (tower exclusion; same bounds, applied uniformly). Running the same machinery on alternative fiber towers (N, M) — each with *its own* derived lattice $Y = Nb - Ma$, its own residue linkages, $|Y| \leq 2NM$, ≤ 5 multiplets, and the corresponding $SU(N)^3$ predicate — yields: $(3, 2)$ floor = 15 Weyl; $(4, 2)$, $(5, 2)$, $(4, 3)$: none within bounds; $(3, 3)$: 36 Weyl. The $(3, 2)$ tower is strictly minimal among probed alternatives; combined with the program's exhaustion theorem (no genuinely chiral record content below stack size three), the fiber dimensions are *minimality-selected*, with the minimality principle itself the named remaining input. One further premise of the comparison is named: for the $(3, 2)$ tower the lattice is derived from a seal whose existence is empirical (Honest negative 3.5), while for the counterfactual towers the analogous baryon-like seal is *stipulated* — the comparison therefore assumes **uniformity** (every tower's ledger seals its own primitive direction), and that assumption stands beside minimality in the input list.

6. Predictions and falsifiers

- **No fractionally charged color singlets** (the quotient structure): a confirmed millicharged particle falsifies the derivation outright.
- **The sixteenth state, honestly graded** — an earlier draft listed "right-handed neutrinos exist" as a forced prediction; review showed that is partially circular: the seam system of Section 4 already includes ν^c-bearing seams among its inputs (bilinears involving ν^c exist only if ν^c does), so the state's *existence* is an input of the seam-existence premise, not an output. What IS derived, conditional on existence and on the *gauged-family premise* (defined here, since it carries this row: the seam system, including the Majorana seal, holds uniformly across the family/generation index): the sixteenth state's only bare mass term is $\nu^c\nu^c$ — Majorana structure is forced, Dirac-only neutrinos are excluded. The corollary "neutrinoless double-beta decay at *some* level" is a **consequence, not a falsifier** — with no rate floor no experiment can refute it, and we list it outside the falsifier set.
- **Baryon-number violation, if seen, is $B-L$-conserving** at leading order (the operator ledger admits $p \to e^+\pi^0$-class channels; $\Delta(B-L) \neq 0$ first at dimension 9).
- Scope of the millicharge falsifier, stated: a confirmed fractionally charged color singlet kills *this derivation* (the $\mathbb{Z}_6$ quotient structure), not the record ontology as a whole — Honest negative 3.5's alternative-seal worlds are exactly the survival route, and we say so rather than let the falsifier appear to cover more than it does.
- Separate registration documents record the program's *quantitative* confrontations: the undressed spectrum-point registration (the note's earlier exponent and band forms are withdrawn within it) and the mechanism paper's four-member dressed menu, two members live against today's data — with tolerances, pre-committed outcome bands, and named resolving experiments.

7. What is input, restated

The complete input list: (1) the generation count (three) is observational; (2) the minimality principle (nature populates the cheapest chiral tower) is named, not derived — its dynamical origin is the program's open dynamics layer; (3) the seal's existence — fermionic carriers in the $(3, 2)$ and $(1, 2)$ composite fibers — is an empirical fact about the ledger (Lemma 3.2, Honest negative 3.5); (4) the screen-$U(1)$ (a surviving continuous charge) is a named input co-equal with the seal (Lemma 3.3); (5) the tower-uniformity premise of the exclusion comparison (Section 5.2); (6) the Yukawa seam existence premise with the **Higgs representation** — the seam carrier is a single weak-doublet, color-singlet scalar, which the graded seam-existence input does not by itself produce and which we name as input here; (7) the gauged-family premise (Section 6); (8) the **global-to-local gauge promotion** — Part II reconstructs each fiber's symmetry as a *global* group (the commutant of exchange) and grades its promotion to a local gauge symmetry with connection and dynamics as an additional input where this paper uses it; the anomaly predicates of Sections 4–5 presuppose the gauged form, so the promotion is counted here as a named input, honoring Part II's grading. Everything else used by the searches is derived with receipts or imported and named (Stone–von Neumann, Schur–Weyl, the Witten condition's topological input). We believe this is the correct standard for claims of this kind, and we commend hostile referees to the audit.

Relation to other derivation programs. The closest structural relative is the noncommutative-geometry program of Connes and Chamseddine [10], which also obtains the Standard-Model fiber structure $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ and hypercharge assignments from an axiomatic selection principle (spectral triples and the spectral action) rather than from phenomenological fitting. The comparison is instructive in both directions: there, the algebra is selected by classification of finite spectral triples and the fermion content sits in a fixed Hilbert space; here, the fibers are selected by a chirality exhaustion theorem plus measured minimality (Theorem 5.2), and the content is the output of an anomaly-constrained search on a derived lattice. Both programs face the same honest residue — a named selection principle (the spectral action's algebra axioms there, the minimality principle here) that the program motivates but does not derive — and both forbid fractionally charged color singlets. The group-theoretic uniqueness literature for the SM gauge quotient [1, 2, 4] is the common backdrop.

Reproducibility

Integer-exact lattice receipts; the seam-system kernel; the 4d index sectors; the floor and tower searches with stated pools and bounds — all regenerate from fixed-seed scripts with bit-identical reruns.

References

[1] L. O'Raifeartaigh, *Group Structure of Gauge Theories*, CUP (1986).

[2] D. Tong, Line operators in the Standard Model, *JHEP* **07** (2017) 104.

[3] C. Q. Geng, R. E. Marshak, *Phys. Rev. D* **39** (1989) 693; R. Foot, G. C. Joshi, H. Lew, R. R. Volkas, *Mod. Phys. Lett. A* **5** (1990) 2721 — hypercharge quantization from anomalies.

[4] J. Baez, J. Huerta, The algebra of grand unified theories, *Bull. AMS* **47** (2010) 483.

[5] E. Witten, An SU(2) anomaly, *Phys. Lett. B* **117** (1982) 324.

[6] K. S. Babu, R. N. Mohapatra, Is there a connection between quantization of electric charge and a Majorana neutrino?, *Phys. Rev. Lett.* **63** (1989) 938; *Phys. Rev. D* **41** (1990) 271.

[7] J. A. Minahan, P. Ramond, R. C. Warner, Comment on anomaly cancellation in the standard model, *Phys. Rev. D* **41** (1990) 715.

[8] H. Neuberger, Exactly massless quarks on the lattice, *Phys. Lett. B* **417** (1998) 141.

[9] P. Hasenfratz, V. Laliena, F. Niedermayer, The index theorem in QCD with a finite cut-off, *Phys. Lett. B* **427** (1998) 125; M. Lüscher, Exact chiral symmetry, topological charge and related topics in lattice QCD, *Nucl. Phys. B Proc. Suppl.* **83** (2000) 34.

[10] A. H. Chamseddine, A. Connes, The spectral action principle, *Commun. Math. Phys.* **186** (1997) 731; A. H. Chamseddine, A. Connes, M. Marcolli, Gravity and the standard model with neutrino mixing, *Adv. Theor. Math. Phys.* **11** (2007) 991.

[11] J. Hucks, Global structure of the standard model, anomalies, and charge quantization, *Phys. Rev. D* **43** (1991) 2709.

[12] B. C. Allanach, B. Gripaios, J. Tooby-Smith, Anomaly cancellation with an extra gauge boson, *Phys. Rev. Lett.* **125** (2020) 161601; Solving local anomaly equations in gauge-rank extensions of the Standard Model, *Phys. Rev. D* **101** (2020) 075015.

Parts I–II: *Quantum theory from sealed records I–II* (same author).